

Overview of Security of Form Applications for Privacy-Minded Groups

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Background

In the era of increasing digital surveillance and data monetization, the simple act of “filling out a form” has become a high-stakes transaction. Many organization, ranging from grassroots activists and non-profits to sensitive research groups, rely on digital forms to collect data from their members. However, these groups often operate under the shadow of significant privacy threats. These threats are not limited to traditional “hackers” (malicious actors); they also include data harvesting by corporations, service providers, and government entities through subpoenas or mass surveillance.

Despite these risks, there is a systemic information gap. The documentation required to perform a rigorous risk analysis is often buried in dense technical white papers, obscured by marketing jargon, or entirely absent. This leaves organizations in a “security blind spot,” forced to choose between the convenience of mainstream tools and the safety of their participants.

Our goal was to bridge this gap. By investigating four form service providers—Google Forms, Jotform, Cryptpad, and Palform—our study evaluates the practical trade-offs between user-friendliness and “zero-trust” security. Our goal is to provide a clear, evidence-based framework that allows organizations to match their specific threat model with the appropriate form service, ensuring that data integrity isn’t sacrificed for the sake of accessibility.

Results

Our investigation revealed a divide between what’s mainstream accessible and verified secure. While Google Forms and Jotform are standardized form platforms across users, they operate on a model of implicit trust where both services utilize a “Manager-Key” system where the provider retains access to user data and explicitly complies with government subpoenas and civil discovery requests annually.

In contrast, Palform emerged as the leader in data protection, exceeding our initial expectations by implementing true End-to-End Encryption via OpenPGP. Unlike mainstream competitors, Palform encrypts responses locally in browser before they reach the server, ensuring that even if the platform was hacked or subpoenaed, the data remains unreadable to everyone except the form owner. While Cryptpad also offers a transparent, open-source alternative with strong anonymity, Palform provided a more modern feature set with a lower learning curve. Although Palform’s free-tier restrictions on response exports can present a barrier for privacy-minded groups, the cost of entry for the next tier is low, at approximately 2 pounds per month. This makes top-tier security an affordable reality for many.

Abstract

Many organizations are interested in collecting information from their members through polls and forms, yet these organizations and their members may have strong interests in keeping their data secure and private: not only from malicious actors, but from anyone who could potentially gain access to the responses, including national governments, corporations, or the form service itself. The documentation needed to perform this kind of risk analysis is often technical, difficult to locate, or nonexistent, making it hard for organizations to understand what services are available and what risks each one poses.

This study serves as an initial investigation into the security and anonymity provided by four form services, Google Forms, Jotform, Cryptpad, and Palform, and the risks each may pose to organizations handling sensitive data.

Using publicly available documentation, we identified key features that a security-conscious form service should provide and referenced third-party security audits to confirm their legitimacy where available. Our analysis was guided by a framework representing the risks organizations may face when handling sensitive data, scoring each service across several categories relevant to both security and practical usability.

We initially found that the most popular form services each presented distinct tradeoffs, suggesting the need for case-by-case evaluation. However, Palform emerged as exceeding expectations in nearly every category investigated, offering strong security guarantees alongside a broad feature set. Its primary limitation is scalability: the free tier restricts response exports, which becomes a significant practical barrier for any organization conducting large-scale data collection.

Although Palform was concluded to be the safest option for the majority of situations, findings show that there is not one complete solution for a secure form service. Organizations should evaluate available providers according to the risk they are willing to take on.

Privacy in Forms Services				
Organizations interested in collecting information from their members through forms are often concerned with the privacy and security of this service. It is important to be aware of who has access to the responses, including other corporations, the government, or the forms service itself. Displayed below is a collection of popular forms services evaluated by their privacy (top half) and features (bottom half).				
Criteria	Google Forms	Jotform	Cryptpad	Palform
EZE Encryption	✗	✗	✓	✓
Anonymity	⚠	✗	✓	✓
Transparency	⚠	✗	✓	✓
Data Ownership	⚠	⚠	✓	✓
Access Control & Authentication	✓	✓	✓	✓
Platform & Accessibility	✓	✓	⚠	✓
File & Media Handling	⚠	✓	✗	✓
Free Version Available	✓	⚠	✓	⚠
This risk analysis was conducted using information from publicly available documentation (by the forms service) and user feedback.				
Legend ✓ Meets all requirements ⚠ Meets some requirements ✗ Does not meet requirements				
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Please explain ...



E2EE (End-to-End Encryption) is the gold standard for protecting your data. **Encryption** transforms your data so only those with the **key** can transform it back (decrypt it). Usually, the web service you are using has that key. With E2EE, the web service doesn't have the key and can't see your data!





Unless an app is funded through donations or subscription fees, it is making money from your data (and loss of privacy). **Freemiums** are less likely to profit from your data – they take payment for additional features or storage space.



You can improve your privacy by accessing the internet through a **VPN (Virtual Private Network)**. A VPN disguises your location meta-data from the websites you visit, but the VPN knows your location and the websites you visit. **Tor**, available through torproject.org, does the same as a VPN but also hides your internet activity from the Tor servers (but breaks some websites).

Use **Google Docs** for low-risk data collected in high volumes.

Use **Jotform** for low to moderate-risk data collection or general surveys.

Use **Cryptpad** for high-risk collection, but in combination with Tor.

Use **Palform** for high-risk data collection.

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Caption: Figure 1.
A one-pager created from information collected during this research. In this format, our findings are made useful in an accessible and understandable way. It is made to appeal to a wide audience, especially organizations interested in collecting sensitive information through forms services. Thi

Methodology

This study used two primary methods to identify and evaluate the security and usability of services offered by multiple providers. First, we conducted a broad review of organizational security and privacy needs by examining both historical security challenges and modern issues affecting current systems. Based on this review, we developed a set of evaluation criteria designed to reflect organizational priorities focused on two major categories: security and usability. These criteria were designed to reflect the priorities and practical concerns of organizations when selecting service providers.

Second, we applied these criteria to a selected group of providers. Each service was evaluated using publicly available documentation, including technical whitepapers, security reports, and user-facing resources. Security assessments focused on provider-published security features, protections, and compliance measures, while usability assessments examined interface functionality, accessibility, and user feedback. This process allowed us to identify both the strengths and limitations of each provider.

To improve clarity and relevance, the team conducted multiple rounds of editing, analysis, and refinement throughout the research process. Research responsibilities were divided among team members, who independently investigated providers before combining findings into a unified evaluation. The group then collaborated through several review cycles to condense categories, improve communication, and determine the most effective use cases and recommendations for organizations.

Future Goals

Future work will expand the evaluation framework to include additional form service providers and more detailed technical analysis. Further research could incorporate hands-on testing, penetration testing methodologies, and direct experimentation with data handling practices rather than relying solely on publicly available documentation.

Additional goals include refining the weighting of security and usability criteria to better reflect the priorities of different organizations, such as nonprofits, activist groups, educational institutions, or healthcare organizations. Future studies may also investigate long-term factors such as regulatory compliance, sustainability of free-tire services, and transparency regarding government or third-party data requests.

Future iterations of this research aim to produce a more accessible decision-making guide that organizations can use to evaluate secure forms according to their needs specific operational and privacy needs.

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